

SANTIAGO PUDAHUEL, Chile

WMO: 855740

Lat: 33.3924S

Lon: 70.7943W

Elev: 481

StdP: 95.68

Time Zone: -4.00 (CHL)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	-0.8	0.4	-4.0	2.9	11.4	-2.7	3.2	8.9	7.4	10.9	6.3	11.6	1.1	0	0.484

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	16.2	32.1	17.0	31.0	16.9	29.9	16.8	18.8	29.1	18.2	28.6	17.6	28.0	6.4	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.2	10.7	22.3	13.9	10.5	22.0	13.1	10.0	21.1	55.2	29.2	53.1	28.6	51.3	28.0	21.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.5	7.8	7.1	-3.0	34.8	1.3	1.1	-3.9	35.6	-4.6	36.3	-5.4	36.9	-6.3	37.7		
			-3.4	19.8	1.2	0.7	-4.3	20.3	-4.9	20.8	-5.6	21.2	-6.5	21.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	15.3	21.5	20.9	19.2	15.4	12.0	9.5	8.9	10.3	12.5	15.1	17.9	20.1
	DBStd	4.96	1.78	1.86	2.00	2.39	2.42	2.37	2.28	2.64	2.51	2.47	2.31	2.05
	HDD10.0	128	0	0	0	1	8	38	48	28	6	0	0	0
	HDD18.3	1429	1	2	14	92	195	266	292	248	174	103	34	7
	CDD10.0	2045	357	306	285	161	72	21	14	38	82	159	237	313
	CDD18.3	306	99	75	41	3	0	0	0	0	0	4	22	62
	CDH23.3	4767	1266	972	699	178	26	1	1	16	49	166	471	922
	CDH26.7	1501	467	339	209	32	2	0	0	1	6	30	120	295
Wind	WSAvg	2.7	3.8	3.5	2.9	2.3	1.8	1.7	1.7	2.0	2.4	2.9	3.5	3.8
Precipitation	PrecAvg	295	1	1	4	18	46	78	57	46	25	10	6	3
	PrecMax	668	6	12	20	105	142	245	185	110	107	56	50	19
	PrecMin	109	0	0	0	0	7	4	9	4	4	0	1	0
	PrecStd	127	1	3	6	25	38	67	39	35	26	15	10	6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.0	33.1	32.3	29.7	26.1	22.1	22.1	25.3	27.2	29.8	31.8	33.0
		MCWB	17.4	17.2	16.6	15.4	13.4	11.9	12.4	13.9	14.5	15.9	15.7	16.9
	2%	DB	32.0	31.4	30.6	27.3	23.2	19.8	19.3	22.1	24.8	27.1	29.5	31.1
		MCWB	17.4	17.2	16.4	14.8	12.7	11.1	11.4	12.6	13.8	14.9	15.9	16.9
	5%	DB	30.8	30.1	29.0	25.5	21.0	17.2	17.0	19.2	22.1	25.1	28.0	29.8
		MCWB	17.2	17.2	16.3	14.4	12.4	10.7	10.4	11.8	13.3	14.4	15.6	16.8
	10%	DB	29.2	29.0	27.4	23.2	18.9	15.1	14.8	16.8	19.8	23.0	26.1	28.2
		MCWB	17.2	17.0	16.0	13.9	12.0	10.4	9.8	10.8	12.5	13.7	15.2	16.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.6	19.6	18.7	17.0	15.0	13.8	13.3	14.9	16.0	17.2	17.9	19.1
		MCDB	30.8	30.2	28.5	25.9	21.3	18.4	20.3	23.0	23.8	26.7	28.4	29.5
	2%	WB	18.7	18.8	17.7	16.0	14.0	12.5	12.1	13.4	14.7	15.8	16.9	18.2
		MCDB	29.3	28.9	27.8	24.2	20.2	16.6	17.5	20.5	22.5	24.9	27.2	28.7
	5%	WB	18.1	18.1	17.1	15.0	13.2	11.6	11.1	12.2	13.7	14.9	16.2	17.5
		MCDB	28.8	28.2	27.0	22.9	19.0	15.6	15.5	17.8	21.0	23.5	26.2	28.0
	10%	WB	17.5	17.5	16.4	14.3	12.3	10.7	10.2	11.3	12.7	14.0	15.4	16.8
		MCDB	28.1	27.4	25.9	21.6	17.6	14.4	14.0	16.0	18.7	21.9	25.1	27.1
Mean Daily Temperature Range	5% DB	MDBR	16.2	16.3	16.5	14.9	13.5	11.9	12.1	12.5	13.1	14.2	15.7	16.2
		MCDBR	18.0	18.1	19.2	19.6	18.7	17.0	17.4	18.5	19.1	18.8	18.7	18.3
		MCWBR	7.1	7.2	7.8	8.9	9.5	9.5	10.0	10.0	9.6	8.5	7.8	7.4
	5% WB	MCDBR	16.3	16.6	17.1	16.5	15.3	13.2	14.4	15.8	17.0	17.0	17.0	16.8
		MCWBR	6.8	7.0	7.5	7.9	8.2	7.9	8.6	8.9	9.0	8.1	7.7	7.3
Clear-Sky Solar Irradiance	taub	0.331	0.332	0.318	0.317	0.309	0.298	0.298	0.309	0.326	0.317	0.309	0.320	
	taud	2.462	2.472	2.502	2.493	2.512	2.569	2.530	2.488	2.419	2.460	2.500	2.479	
	Ebn at Noon	1006	985	963	908	861	849	866	903	939	989	1023	1021	
	Edh at Noon	118	113	102	92	80	71	77	91	109	113	113	117	
All-Sky Solar Radiation	RadAvg	8.53	7.67	6.28	4.38	2.81	2.16	2.38	3.08	4.45	5.97	7.61	8.55	
	RadStd	0.42	0.28	0.34	0.42	0.32	0.30	0.25	0.37	0.44	0.45	0.44	0.32	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	+0.48	+0.53	N/A	+0.64	N/A	N/A	N/A	-84	+187	+105
(47)	Regional (3 neighbors)	+0.38	N/A	-1.17	+0.58	N/A	N/A	N/A	N/A	+137	+93

Nomenclature: See separate page